

Project Cinema

Product Requirements Document

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1. Vision

A personalised movie recommendation system where users rate films, write reviews, and receive intelligent recommendations that improve over time. An LLM layer handles natural language input and explains every recommendation in plain English. Built as a portfolio project, now deployed as a live web application.

2. Problem Statement

Existing recommenders (Netflix, Letterboxd) are black boxes — they tell you what to watch but never why. Users have no way to express nuanced preferences in natural language, and recommendations don't visibly improve as the user engages more. This project solves both: transparent, explainable recommendations driven by real user behaviour and conversational input.

3. Target Users

User Type	Description
New user	No history. Onboarded via genre/mood preferences. Gets content-based recs immediately.
Returning user	Has rated 5+ movies. Gets collaborative filtering recs that improve over time.
Power user	Has rated 20+ movies and written reviews. Gets hybrid recs + full LLM explanations.

4. Core Features

4.1 Rating System — 5-Tier Visual Scale

Each tier has a distinct visual identity — icon, colour, and label. No stars, no numbers.

Tier	Icon	Colour	Meaning
Peak Cinema	Flame	#C8821A	Absolute best — life-changing
Masterpiece	Diamond	#D4A017	Exceptional craft
Great Watch	Circle	#2D9CDB	Solid, enjoyable

Mid	Dash	#8A7560	Forgettable
Skip	Cross	#8B4513	Not worth your time

- Tiers displayed as pill badges with icon + label
- On rating, selected tier animates with a brief pulse/glow
- Tier history shown as a visual distribution bar on user profile

4.2 Genre Visual System

Each genre has a unique icon and colour tag — displayed as compact icon+label chips on movie cards.

Genre	Icon	Colour
Action	Lightning	#E8712A
Drama	Masks	#1A5A8A
Thriller	Eye	#8B0000
Comedy	Sun	#F5C518
Sci-Fi	Diamond	#00B4D8
Horror	Moon	#4A0E8F
Romance	Heart	#E8436A
Animation	Star	#7CB518

4.3 Recommendation Engine (ML Core)

Three-layer hybrid pipeline:

Layer	Model	Purpose
Content-based	Random Forest	New users, cold-start. Uses movie metadata + genre features.
Collaborative filtering	SVD (Matrix Factorization)	Warm users. Learns from rating patterns across all users.
Hybrid fusion	Weighted ensemble	Blends CF + content scores. CF weight increases as user rates more.

4.4 LLM Layer (Claude API)

Two roles:

- **Input handler** — User types natural language preferences. LLM extracts structured intent (genres, mood, runtime, tone) and passes it to the recommendation engine as a feature query.

- **Output explainer** — For every recommendation, LLM generates a 2-3 sentence explanation referencing the user's top-rated films and taste profile.

4.5 Explainability — Top 3 Recommendation Cards

The XAI output is presented as 3 cinematic recommendation cards displayed side by side. Each card contains:

- Movie poster via TMDB API (fetched using links.csv TMDB IDs)
- Title + year + genre chips
- Predicted tier badge (with icon + colour)
- A single bold match reason headline
- 2-3 sentence LLM explanation below the headline
- Confidence arc indicator unique to each card
- "Rate it" + "Watchlist" actions

Card design: warm parchment background (Dune palette), faint warm glow on hover, top pick slightly larger than 2 and 3, staggered fade-in animation.

4.6 TMDB Integration — Also Worth Exploring

Below the ML recommendations, 3 live TMDB picks are shown in an 'Also Worth Exploring' section. Seeded using the user's top-rated movie titles to ensure relevance, filtered to exclude already-rated movies. A discover fallback uses genre-based TMDB discovery when search yields insufficient results. Available across all three tabs: For You, By Genre, Ask Claude.

4.7 Review System

- Short-form text reviews (optional, attached to a rating)
- Stored in SQLite with tier badge + username
- Reviews feed into sentiment analysis in Phase 3

4.8 User Profile

- Taste fingerprint: top genres (shown as icon chips), favourite tier distribution
- Rating history with tier badges and release year
- Avatar upload (base64, stored in SQLite)
- Recommendation feed that updates as user rates more

4.9 Search

Unified movie search combining MovieLens local data and TMDB API results. Returns posters, genre chips, and release year. Used to rate any movie directly from the home page.

5. Data Architecture

5.1 Training Data

Source: MovieLens ml-latest-small (100k ratings, 9,742 movies, 610 users)

Files: ratings.csv, movies.csv, tags.csv, links.csv — committed to GitHub repo.

5.2 User Data

New ratings and reviews stored in SQLite (lumiere.db). History-derived genre preferences used to personalise recommendations on every request. Already-rated movies excluded from future recommendation results.

5.3 Feature Matrix

Feature	Source	Type	Status
user_avg_rating	User history	Numeric	Live
user_rating_count	User history	Numeric	Live
user_rating_std	User history	Numeric	Live
avg_rating	MovieLens aggregate	Numeric	Live
rating_count	MovieLens aggregate	Numeric	Live
release_year	movies.csv	Numeric	Live
genre flags (8)	movies.csv	Binary	Live
all_tags	tags.csv aggregated	Text	Phase 3
sentiment_score	User reviews	Numeric	Phase 3

6. Technical Stack

Layer	Technology
ML models	scikit-learn 1.3.2 (LR, RF), scipy 1.11.4 (SVD)
LLM	Anthropic Claude API — claude-sonnet-4-6 (anthropic==0.40.0)
Explainability	SHAP TreeExplainer (local); rule-based fallback on Render free tier
Backend	Python 3.11.4 + FastAPI 0.110.0 + Uvicorn
Database	SQLite (lumiere.db) via raw sqlite3
Frontend	React 18 + Vite + inline styles (Dune design system)
Deployment	Render — lumiere-api (Python) + lumiere-frontend (Static)
Keep-alive	UptimeRobot — pings /health every 5 min to prevent cold start

Poster API

TMDB API v3 (Bearer token auth, w300 images)

7. GitHub Repository Structure

Repository: github.com/Casual-Kalotra-328/project-cinema

```
project-cinema/
  README.md
  requirements.txt
  render.yaml
  .python-version          (3.11.4 -- pinned for Render)
  .env.example
  .gitignore
  data/
    ml-latest-small/       (ratings, movies, tags, links CSVs)
  notebooks/
    cinema_recommender.ipynb
  ml/
    features.py             (feature engineering)
    train.py                (model training pipeline)
    predict.py              (inference + top-N recommendations)
    evaluate.py             (metrics + SHAP)
    llm.py                  (Claude API integration)
    models/                 (rf, lr, svd_matrix, scaler .pkl files)
  api/
    main.py                 (FastAPI app)
    db/
      database.py           (SQLite -- users, ratings, reviews)
  frontend/
    src/
      components/
        TierBadge.jsx
        GenreChip.jsx
        RecommendationCard.jsx
      styles/dune.css
      App.jsx
  docs/
    PRD.md
    CHANGELOG.md
```

8. Phased Delivery Plan

Phase 1 — ML Core + Portfolio Demo [Complete]

- Clean training pipeline on ml-latest-small
- LR + RF + SVD models trained and evaluated
- SHAP explainability working (local)

- Top 3 recommendation cards with LLM explanations (Claude API)
- Genre chip + Tier badge visual system
- Interactive React demo
- Notebook published in /notebooks

Phase 2 — Full Web App [Complete]

- FastAPI backend with recommendation, rating, search endpoints
- User accounts with name, email, avatar
- Ratings + reviews stored in SQLite
- React frontend with full Dune design system deployed on Render
- TMDb poster integration + Also Worth Exploring section (all 3 tabs)
- DB-aware recommendations — excludes already-rated movies
- History-derived genre preferences for personalisation
- SHAP disabled on Render free tier (512MB limit); rule-based fallback used
- Note: PostgreSQL migration deferred to Phase 3

Phase 3 — Real Product

- Sentiment analysis on reviews as ML feature
- Dark mode
- Social layer — see what friends rated
- PostgreSQL migration from SQLite
- Mobile browser compatibility fix
- SHAP re-enabled on higher-memory instance

Phase 4 — Mobile

- React Native app (iOS + Android)

9. Success Metrics

Metric	Phase 1 Target	Phase 2 Target	Actual
RF classification accuracy	> 40% (5-class)	> 50% with user data	74.39% (LR)
SVD RMSE	< 1.0	< 0.85	< 1.0
LLM explanation quality	Coherent + relevant	User rates useful > 70%	Live
Cold-start coverage	100% new users	--	100%
Deployment	Local	Render	Live

10. What This Demonstrates (Resume Value)

Skill	Where it shows
End-to-end ML pipeline	Data -> features -> training -> evaluation -> deployment
Collaborative filtering	SVD matrix factorization on 100k ratings
LLMs in production	Claude API for input parsing + output explanation
Explainable AI	SHAP values, rule-based fallback, top-3 card presentation
Real dataset	MovieLens (industry benchmark)
Product design	Design system, tier/genre visual language, Dune palette
Full-stack	FastAPI + React + SQLite + Render deployment
API integration	TMDB REST API, Anthropic Claude API
NLP	Tag aggregation, LLM intent parsing, review sentiment (Phase 3)